

The $|\mathfrak{B}_2|^2$ row

N_1 (three ρ) and N_2 (two ρ) — transverse divergences, then the p^+ integral

$$S \equiv p^+ + k^+, \quad A \equiv (x - z)^2, \quad B \equiv (x - w)^2, \quad C \equiv (z - w)^2, \quad A' \equiv (x' - z)^2, \quad B' \equiv (x' - w)^2, \quad D = p^+ A + k^+ B$$

Summary. Both prefactors are **correct** (§1), but one piece of $|\mathfrak{B}_2|^2$ is missing and two measures are wrong. On divergences: **there is no ultraviolet divergence in either N_1 or N_2** — all 26 collapsing regions of N_2 and all 57 of N_1 converge (§2). The transverse divergence that is there is **infrared**, at large $|z|$, and it has a clean mechanism: the one- ρ (gluon-exchange) part of $\mathfrak{B}_2$ degenerates at large $|z|$ into **exactly** the two- ρ part's kernel, with the same prefactor up to a factor i (§3). That ties the z -divergences of N_1 and N_2 to the 4ρ term you left out — they are not independent. The p^+ integrals then go through in closed form: three masters for N_1 (§4), six for N_2 (§5), all verified against quadrature.

1. The decomposition

Split the vertex as $\mathfrak{B}_2 = \mathfrak{B}^{(1)} + \mathfrak{B}^{(2)}$ with

$$\mathfrak{B}^{(1)} : \frac{ig^2 \sqrt{p^+ k^+}}{4\pi^2 S} \quad (\text{one } \rho), \quad \mathfrak{B}^{(2)} : -\frac{g^2}{8\pi^2 \sqrt{p^+ k^+}} \quad (\text{two } \rho).$$

piece	ρ 's	prefactor from $\frac{4}{(2\pi)^3} \mathfrak{B} \mathfrak{B}^\dagger$	yours
$(1\rho) \times (1\rho)$	2	$\frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{p^+ k^+}{S^2}$	N_2 ✓
$(1\rho) \times (2\rho)$	3	$-\frac{1}{(2\pi)^3} \frac{ig^4}{8\pi^4} \cdot \frac{1}{S}$	N_1 ✓
$(2\rho) \times (2\rho)$	4	$\frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{p^+ k^+}$	missing

In N_1 the $\sqrt{p^+ k^+}$ and $1/\sqrt{p^+ k^+}$ cancel, and the vertex's overall $\frac{1}{S}$ distributes as

$$\frac{1}{S} \left[\frac{\delta^{ij}(A-B)}{2C} + \frac{S}{k^+}(\dots) + \frac{S}{p^+}(\dots) \right] = \frac{\delta^{ij}(A-B)}{2SC} + \frac{1}{k^+}(\dots) + \frac{1}{p^+}(\dots),$$

which is your N_1 bracket exactly. ✓

Three things to fix.

1. The 4ρ **term is absent**. It is a pure product of four WW kernels, $\frac{(y-w)^i(x-z)^j(y'-w')^i(x'-z)^j}{(y-w)^2(x-z)^2(y'-w')^2(x'-z)^2}$, and it carries $\frac{1}{p^+ k^+}$ — a *double* soft factor. §3 shows it is not optional: it is exactly what the z -divergences of N_1 and N_2 run into.
2. N_2 's measure reads $\int_{z,w,w'}$ but x and x' appear in the integrand — it should be $\int_{x,x',z,w,w'}$.
3. N_2 's two brackets both use i, j contracted; keep the slot order W_{ji} consistent (j with $(x - z)$, i with $(x - w)$) or the cross terms transpose.

2. No ultraviolet divergence, in either piece

Write the bracket as a matrix $W_{ji}(x, z, w)$ (the S -multiplied form), so

$$N_1 \propto \frac{(x' - w')^i (y - z)^j}{(x' - w')^2 (y - z)^2} \frac{W_{ji}(x, z, w)}{S D}, \quad N_2 \propto \frac{p^+ k^+}{S^2} \frac{W_{ji}(x, z, w) W_{ji}(x', z, w')}{D D'}.$$

Every subset of the integration points was collapsed at scale ρ against the measure $\rho^{2(n-1)} \frac{d\rho}{\rho}$. Representative rows (full tables in the results file):

piece	region	$\rho = 10^{-2}$	10^{-4}	10^{-6}	power
N_2	$\{x, z, w\} \ (D \rightarrow 0)$	9.325e-6	9.325e-10	9.325e-14	ρ^2
N_2	$\{z, w, w'\}$	3.958e-5	3.958e-9	3.958e-13	ρ^2
N_1	$\{z, w\}$	1.538e-2	1.538e-4	1.538e-6	ρ^1
N_1	$\{x, z, w\} \ (D \rightarrow 0)$	3.263e-5	3.263e-9	3.263e-13	ρ^2
N_1	$\{y, z\}$	1.464e-2	1.461e-4	1.461e-6	ρ^1

The one region worth naming is $x, z, w \rightarrow$ a point, where the propagator denominator $D = p^+ A + k^+ B$ vanishes like ρ^2 . But the bracket stays $O(1)$ there, so the integrand is ρ^{-2} against a three-point measure $\rho^3 d\rho$ — convergent, with room to spare. **All 26 regions of N_2 and all 57 of N_1 fall. There is no UV.**

3. The real transverse divergence: large $|z|$, and what it means

Send $|z| \rightarrow \infty$ with everything else fixed. In W_{ji} the $\frac{S}{k^+}$ structure grows like $|z|$ (through $-\frac{(x-z)^j(x-w)^i}{2(x-w)^2}$) while $D \sim p^+ |z|^2$, and the limit is clean:

$$\frac{W_{ji}(x, z, w)}{S D(x, z, w)} \xrightarrow{|z| \rightarrow \infty} -\frac{1}{2p^+ k^+} \frac{(x-w)^i}{(x-w)^2} \frac{(x-z)^j}{(x-z)^2}.$$

Checked pointwise (not angular-averaged — the angular average of a single WW kernel vanishes), component by component:

$ z $	10^2	10^3	10^4	10^5	10^6
exact/limit, [0, 0]	1.014187	1.001461	1.000146	1.000015	1.000001
exact/limit, [1, 0]	1.039259	1.003899	1.000390	1.000039	1.000004

3.1 The one- ρ vertex becomes the two- ρ vertex

Feed that limit back into $\mathfrak{B}^{(1)}$ and compare with $\mathfrak{B}^{(2)}$:

$$\begin{aligned} \mathfrak{B}^{(1)} \Big|_{|z| \rightarrow \infty} &= -\frac{ig^2}{8\pi^2 \sqrt{p^+ k^+}} \int_x f^{cba} \rho^a(x) \frac{(x-w)^i (x-z)^j}{(x-w)^2 (x-z)^2}, \\ \mathfrak{B}^{(2)} &= -\frac{g^2}{8\pi^2 \sqrt{p^+ k^+}} \int_{x,y} \{\rho^c(y), \rho^b(x)\} \frac{(y-w)^i (x-z)^j}{(y-w)^2 (x-z)^2}. \end{aligned}$$

Same kernel, same $\frac{g^2}{8\pi^2 \sqrt{p^+ k^+}}$ — the ratio of prefactors is exactly i . They differ only in the colour/source content: $-i f^{cba} \rho^a(x)$ with both kernels at the *same* point, against $-\{\rho^c(y), \rho^b(x)\}$ with the kernels at *independent* points. Physically: at large $|z|$ the exchanged-gluon graph stops resolving the exchange and becomes a product of two independent Weizsäcker–Williams emissions.

3.2 Consequence — a logarithmic infrared divergence, with known coefficient

So both pieces have $\langle \text{integrand} \rangle |z|^2 \rightarrow \text{const}$, i.e. a log divergence at the large- $|z|$ end. The constants are predicted by §3 and confirmed to nine decimals:

$ z $	10^2	10^3	10^4	10^5	10^6	predicted
N_1	-0.456844414	-0.456891576	-0.456892049	-0.456892053	-0.456892053	-0.456892053
N_2	0.228435566	0.228445922	0.228446026	0.228446027	0.228446027	0.228446027

$$N_1 : -\frac{1}{2p^+ k^+} \frac{(x-w) \cdot (x' - w')}{(x-w)^2 (x' - w')^2}, \quad N_2 : +\frac{1}{4p^+ k^+} \frac{(x-w) \cdot (x' - w')}{(x-w)^2 (x' - w')^2}.$$

How to handle it. It is not UV, so no short-distance counterterm applies. Two statements follow, and the second is the important one.

- **Regulate it the same way as the previous row:** $\int d^{2-2\epsilon} z$ with $\epsilon > 0$ damps the large- $|z|$ end and turns the log into a $\frac{1}{\epsilon_{\text{IR}}}$ pole. Nothing new is needed.
- **Do not treat N_1 and N_2 separately at that end.** Their z -tails carry the same transverse structure and the same $\frac{1}{p^+k^+}$ as the 4ρ term you dropped — which is the term built from precisely that kernel squared. The three pieces have to be combined before the z integral is regulated, or the $\frac{1}{\epsilon}$'s will not come out right. Restore the 4ρ term first.

4. The p^+ integral for N_1

All the p^+ dependence is $\frac{1}{D}$ times the three bracket weights $\frac{1}{2S}, \frac{1}{p^+}, \frac{1}{k^+}$. Three masters, $P_V = \vee - k^+$:

$$\begin{aligned}\mathcal{J}_1 &= \int_{\Lambda}^{P_V} \frac{dp^+}{p^+ D} = \frac{1}{k^+ B} \log \frac{P_V k^+ B}{\Lambda (P_V A + k^+ B)} \xrightarrow{\vee \rightarrow \infty} \frac{1}{k^+ B} \log \frac{k^+ B}{\Lambda A}, \\ \mathcal{J}_2 &= \int_{\Lambda}^{P_V} \frac{dp^+}{D} = \frac{1}{A} \log \frac{P_V A + k^+ B}{\Lambda A + k^+ B} \rightarrow \frac{1}{A} \log \frac{\vee A}{k^+ B}, \\ \mathcal{J}_3 &= \int_{\Lambda}^{P_V} \frac{dp^+}{(p^+ + k^+) D} = \frac{1}{k^+ (B - A)} \log \frac{\vee B}{P_V A + k^+ B} \rightarrow \frac{1}{k^+ (B - A)} \log \frac{B}{A}.\end{aligned}$$

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k+=1, A=0.8, B=1.7, Lambda=1e-8, V=1e7
J1: quad 11.279089611 closed 11.279089608 diff 2.8e-09
J2: quad 19.205404945 closed 19.205404945 diff 1.4e-12
J3: quad 0.837524094 closed 0.837524100 diff 5.9e-09
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The δ^{ij} term collapses exactly. Its weight is $\frac{A-B}{2C} \mathcal{J}_3$, and the $(A - B)$ cancels the $(B - A)$:

$$\frac{A - B}{2C} \cdot \frac{\log(B/A)}{k^+(B - A)} = -\frac{1}{2k^+ C} \log \frac{B}{A} \implies -\frac{\delta^{ij}}{2k^+(z - w)^2} \log \frac{(x - w)^2}{(x - z)^2}.$$

No $\vee$, no Λ — numerically exact to 12 digits.

So, with $\mathcal{W}_p^{ij}$ and $\mathcal{W}_k^{ij}$ the two p^+ -independent tensors:

$$\frac{d^3 N_1}{d^3 k} = -\frac{1}{(2\pi)^3} \frac{ig^4}{8\pi^4} \frac{1}{2\pi} \int_{x,y,z,w,w'} e^{-ik(w'-w)} \frac{(x' - w')^i (y - z)^j}{(x' - w')^2 (y - z)^2} \left\{ -\frac{\delta^{ij} \log \frac{B}{A}}{2k^+ C} + \mathcal{W}_p^{ij} \mathcal{J}_1 + \frac{\mathcal{W}_k^{ij}}{k^+} \mathcal{J}_2 \right\}$$

plus the second, $w \leftrightarrow w'$ term with the opposite sign. The $\log \frac{1}{\Lambda}$ sits only in $\mathcal{J}_1$; the $\log \vee$ only in $\mathcal{J}_2$.

5. The p^+ integral for N_2

Writing $W = T_1 + \frac{S}{k^+} T_2 + \frac{S}{p^+} T_3$ with $T_{1,2,3}$ all p^+ -independent, the product $\frac{p^+ k^+}{S^2} W_{ji} W'_{ji}$ gives exactly six longitudinal weights:

tensor structure	weight $h(p^+)$	behaviour
$T_1 T'_1$	$\frac{p^+ k^+}{S^2}$	safe both ends
$T_1 T'_2 + T_2 T'_1$	$\frac{p^+}{S}$	safe both ends
$T_1 T'_3 + T_3 T'_1$	$\frac{k^+}{S}$	safe both ends
$T_2 T'_2$	$\frac{p^+}{k^+}$	$\log \vee$
$T_3 T'_3$	$\frac{k^+}{p^+}$	rapidity $\log \frac{1}{\Lambda}$
$T_2 T'_3 + T_3 T'_2$	1	safe both ends

With $D D' = A A' (p^+ + \alpha)(p^+ + \alpha')$, $\alpha = \frac{k^+ B}{A}$, $\alpha' = \frac{k^+ B'}{A'}$, every master is elementary:

$$\mathcal{M}[h] = \frac{1}{AA'} \int_{\Lambda}^{P_V} \frac{h(p^+) dp^+}{(p^+ + \alpha)(p^+ + \alpha')},$$

$$\mathcal{M}[1] = \frac{1}{AA'} \frac{1}{\alpha' - \alpha} \log \frac{p^+ + \alpha}{p^+ + \alpha'}, \quad \mathcal{M}\left[\frac{p^+}{k^+}\right] = \frac{1}{k^+ AA'(\alpha' - \alpha)} \left[\alpha' \log(p^+ + \alpha') - \alpha \log(p^+ + \alpha) \right],$$

$$\mathcal{M}\left[\frac{k^+}{p^+}\right] = \frac{k^+}{AA'} \left[\frac{\log p^+}{\alpha \alpha'} + \frac{\log(p^+ + \alpha)}{\alpha(\alpha - \alpha')} + \frac{\log(p^+ + \alpha')}{\alpha'(\alpha' - \alpha)} \right],$$

and the three with the extra pole at $p^+ = -k^+$ follow from the residues $\frac{1}{(\alpha - k^+)(\alpha' - k^+)}$ etc., with a double pole for $\frac{p^+ k^+}{s^2}$. All six against quadrature, $\Lambda = 10^{-8}$, $\vee = 10^7$:

structure	quadrature	closed form	abs.diff
T2T3' h=1	0.882636709	0.882636709	0.0e+00
T2T2' h=p/k	14.366017631	14.366017631	1.4e-12
T3T3' h=k/p	16.886103148	16.886103148	3.7e-13
T1T3' h=k/(p+k)	0.442719469	0.442719469	1.2e-15
T1T2' h=p/(p+k)	0.439917240	0.439917240	1.8e-15
T1T1' h=pk/(p+k)^2	0.144304923	0.144304923	4.9e-15

And the attribution of the two logarithms, measured as slopes:

structure	$\partial/\partial \log \frac{1}{\Lambda}$	predicted	$\partial/\partial \log \vee$	predicted
$T_3 T_3' (k^+/p^+)$	0.980391876	$\frac{1}{k^+ BB'} = 0.980392157$	0.000000000	0
$T_2 T_2' (p^+/k^+)$	0.000000000	0	0.961535762	$\frac{1}{k^+ AA'} = 0.961538462$
the other four	$\leq 10^{-7}$	0	$\leq 10^{-6}$	0

$$\text{rapidity log: } \frac{1}{k^+(x-w)^2(x'-w')^2} \log \frac{1}{\Lambda} \text{ (only from } T_3 T_3')$$

$$\text{log } \vee \text{ residue: } \frac{1}{k^+(x-z)^2(x'-z')^2} \text{ (only from } T_2 T_2')$$

6. Where this leaves you

- **No UV in either piece**, so no transverse counterterm and no $\overline{\text{MS}}$ subtraction is needed before the p^+ integral.
- **The z integral is IR log divergent in both**, with the coefficients of §3.2. Regulate in $d = 2 - 2\epsilon$, $\epsilon > 0$, as in the previous row — **but restore the 4ρ term first**, because the tails of N_1 , N_2 and N_4 are all built from the same $\frac{(x-w)^i(x-z)^j}{(x-w)^2(x-z)^2}$ kernel.
- **The p^+ integrals are done**: closed forms in §4 and §5, all verified. The rapidity logarithm is isolated in one structure in each piece ($\mathcal{J}_1$ for N_1 , $T_3 T_3'$ for N_2), so the $[\frac{1}{p^+}]_+$ prescription applies to that structure alone — exactly as in the previous rows.
- **The $\log \vee$ is real and not a rapidity log** ($\mathcal{J}_2$, $T_2 T_2'$). It comes from the large- p^+ end where the eikonal treatment breaks down; it will not be absorbed by evolution and should be checked against the kinematic bound.

zint/src/row9_pref.py (§1), row9_uv.py (§2), row9_zfar.py, row9_degen.py (§3), row9_n1masters.py (§4), row9_masters.py (§5) — every number above.